Two Phase Locking Exercises

For each schedule, write the two-phase locking steps as we discussed in class.

Show the resulting dependence graph of the conflict serializable schedule it produces.

Schedule 1

$W_1(A), R_2(A), W_1(B), R_2(B)$

Schedule 2

$W_1(A), R_2(A), W_2(B), R_1(B)$

Schedule 3

$R_1(A), R_2(A), W_3(A), R_1(B)$

Schedule 4

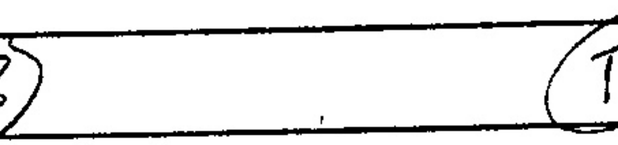
$R_1(A), W_2(A), W_1(A)$

Schedule 5

$W_1(A), R_2(A), W_2(B), R_3(B), W_3(C), R_1(C)$

TWO-PHASE LOCKING EXERCISES: BEFORE 2PL:

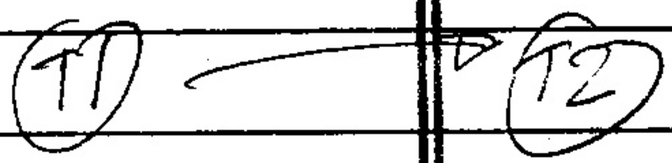
① W1(A), R2(A), W1(B), R2(B)



T1	T2	WQ
XL(A) - G		R2(A)
W1(A)		
	SL(A) - W	
XL(B) - G		
W1(B)		
U(A)		
	SL(A) - G	
	R2(A)	
	SL(B) - G	
	R2(B)	
	U(A), U(B)	

SCHEDULE AFTER 2PL

W1(A), W1(B), R2(A), R2(B)



T1 → T2

(2) W1(A), R2(A), W2(B), R1(B)

T1	T2	W6
~~XL(A) G~~		~~R2(A)~~
W1(A)		~~W2(B)~~
	SL(A) - W	
SL(B) - G		
R1(B)		
U(A), U(B)		
	SL(A) - G	
	R(A)	
	XL(B) - G	
	W(B)	
	U(A), U(B)	

SCHEDULE AFTER 2PL:

W1(A), R1(B), R2(A), W2(B)

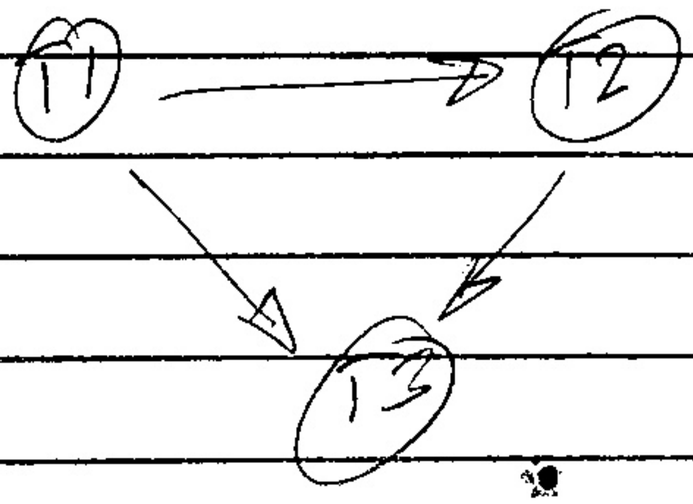
(T1) → (T2) ~~T1 - T2~~

(3) R1(A), R2(A), W3(A), R1(B)

T1	T2	T3	WQ
SL(A)-G			W3(A)
R1(A)			
	SL(A)-G		
	R2(A)		
	U(A)		
		XL(A)-W	
SL(B)-G			
R1(B)			
U(A), U(B)			
		XL(A)-G	
		W3(A)	
		U(A)	

SCHEDULE AFTER 2PL

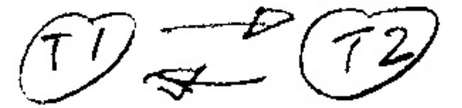
R1(A), R2(A), R1(B), W3(A)



T1 → T2 → T3

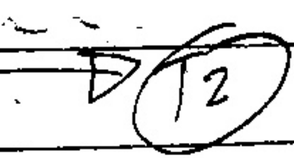
T1 → T3

(4) R1(A), W2(A), W1(A)



T1 → T2 → T1

T1	T2	WQ
SL(A) - G		W2(A)
R1(A)		
	XL(A) - W	
		SCHEDULE AFTER 2PL:
XL(A) - G		R1(A), W1(A), W2(A)
W1(A)		
U(A)		
	XL(A) - G	
	W2(A)	
	U(A)	



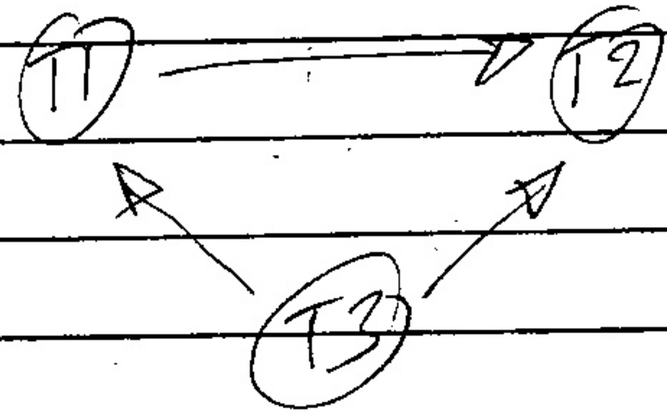
T1 → T2

(5) W1(A), R2(A), W2(B), R3(B), W3(C), R1(C)

T1	T2	T3	WQ
XL(A) - G			R2(A)
W1(A)			W2(B)
	SL(A) - W		
		SL(B) - G	
		R3(B)	
		XL(C) - G	
		W3(C)	
		U(B), U(C)	
SL(C) - G			
R1(C)			
U(A), U(C)			
	SL(A) - G		
	R2(A)		
	XL(B) - G		
	W2(B)		
	U(A), U(B)		

SCHEDULE AFTER RECOVERY:

W1(A), R3(B), W3(C), R1(C), R2(A), W2(B)



T1 → T2
T3 → T2
T3 → T1